Welcome

ADVANCED PROJECT MANAGEMENT IN AI

Lesson 5 Session 1

Agile and Team Management in AI Projects:

Agile Methodologies

Welcome

课程：
人工智能高级项目管理

第5课 第1节

人工智能项目的敏捷与团队管理：敏捷方法论

LESSON 5.1 – Agile Methodologies in AI Projects



第5.1课 – 人工智能项目中的敏捷方法论



Learning Objectives: Lesson 5.1

- Explain why traditional project management approaches struggle with AI development
- Distinguish between Waterfall and Agile approaches in the context of AI projects
- Understand iterative experimentation cycles used in machine learning development
- Describe principles of Lean AI development
- Evaluate when hybrid project management approaches are appropriate for AI initiatives

学习目标：第5.1课

- 解释为什么传统的项目管理方法难以应对人工智能开发
- 区分瀑布式方法与敏捷方法在人工智能项目背景下的应用
- 理解机器学习开发中使用的迭代实验周期
- 描述精益人工智能开发的原则
- 评估混合项目管理方法何时适用于人工智能项目

Why Agile Matters for AI Project Managers

From a project management perspective, AI initiatives introduce high levels of uncertainty that challenge traditional planning models.

Project managers must deal with:

- Unpredictable model outcomes during development
- Data availability and quality challenges
- Iterative experimentation rather than linear development
- Frequent adjustments to problem definitions
- Rapid evolution of tools, frameworks, and algorithms

Agile methodologies provide flexibility and adaptive planning, enabling teams to learn from experimentation and adjust project direction accordingly.

(Project Management Institute, 2021)

为什么敏捷对人工智能项目经理很重要

从项目管理的角度来看，人工智能项目引入了高度的不确定性，这对传统的规划模式构成了挑战。

项目经理必须应对：

- 开发过程中不可预测的模型结果
- 数据的可用性和质量问题
- 迭代式实验而非线性开发
- 对问题定义的频繁调整
- 工具、框架和算法的快速演进

敏捷方法论提供了灵活性和适应性规划，使团队能够从实验中学习并相应调整项目方向。

（项目管理协会，2021年）

Traditional Waterfall Approach

The Waterfall model follows a sequential process where each phase must be completed before the next begins.

Typical stages include:

- Requirements definition
- System design
- Development
- Testing
- Deployment

This approach works well when:

- Requirements are stable
- Technology is well understood
- Outputs are predictable

However, these assumptions rarely hold true for AI projects.

传统瀑布式方法

瀑布式模型遵循一个顺序流程，每个阶段必须完成后才能开始下一个阶段。

典型阶段包括：

- 需求定义
- 系统设计
- 开发
- 测试
- 部署

这种方法在以下情况下效果良好：

- 需求稳定
- 技术已充分理解
- 输出可预测

然而，这些假设在人工智能项目中很少成立。

Why Waterfall Often Fails in AI Projects

AI development involves significant experimentation and uncertainty, which conflicts with the rigid structure of Waterfall planning.

Common challenges include:

- Model accuracy cannot be predicted in advance
- Data quality issues emerge during development
- Feature engineering requires repeated iterations
- Business objectives evolve as insights emerge
- Model validation may require redesign of earlier stages

As a result, strictly sequential project structures often lead to delays, rework, and scope instability.

为什么瀑布式方法在人工智能项目中常常失败

人工智能开发涉及大量的实验和不确定性，这与瀑布式规划的刚性结构相冲突。

常见的挑战包括：

- 模型准确性无法预先预测
- 数据质量问题在开发过程中浮现
- 特征工程需要反复迭代
- 随着见解的出现，业务目标也会演变
- 模型验证可能需要重新设计早期阶段

因此，严格的顺序项目结构常常导致延误、返工和范围不稳定。

Agile Methodology: Core Principles

Agile project management emphasizes adaptability, collaboration, and incremental progress.

Key principles include:

- Iterative development cycles
- Frequent feedback and testing
- Cross-functional team collaboration
- Continuous improvement
- Flexibility in responding to change

These principles align closely with the experimental nature of machine learning development.

(Beck et al., 2001 – Agile Manifesto)

敏捷方法论：核心原则

敏捷项目管理强调适应性、协作和渐进式进展。

关键原则包括：

- 迭代开发周期
- 频繁的反馈和测试
- 跨职能团队协作
- 持续改进
- 响应变化的灵活性

这些原则与机器学习开发的实验性质高度契合。

（Beck 等人，2001 年 – 敏捷宣言）

Iterative Experimentation Cycles in AI

Machine learning systems are built through repeated experimentation cycles.

A typical AI experimentation loop includes:

1. Data preparation
2. Feature engineering
3. Model training
4. Model evaluation
5. Performance improvement

Each cycle generates new insights that guide the next iteration of model development.

This iterative structure fits naturally with Agile sprint-based workflows.

人工智能中的迭代实验周期

机器学习系统是通过反复的实验周期构建的。

典型的人工智能实验循环包括：

1. 数据准备
2. 特征工程
3. 模型训练
4. 模型评估
5. 性能改进

每个周期都会产生新的见解，指导下一轮模型开发。

这种迭代结构天然地契合基于敏捷冲刺的工作流程。

The Role of Feedback in AI Development

Frequent feedback is essential in AI projects because:

- Model performance must be continuously evaluated
- Stakeholders must validate whether predictions are useful
- Data issues must be identified early
- Ethical or bias concerns may emerge during testing

Agile practices enable teams to incorporate feedback quickly, reducing the risk of building models that fail to meet business needs.

反馈在人工智能开发中的作用

频繁的反馈在人工智能项目中至关重要，因为：

- 必须持续评估模型性能
- 利益相关者必须验证预测是否有效
- 数据问题必须尽早发现
- 在测试过程中可能出现伦理或偏见问题

敏捷实践使团队能够快速整合反馈，降低构建无法满足业务需求的模型的风险。

Lean AI Development

Lean principles emphasize maximizing value while minimizing waste in development processes.

In AI projects, Lean development focuses on:

- Rapid prototyping of models
- Early testing with small datasets
- Eliminating unnecessary complexity
- Prioritizing features that deliver business value

This approach helps organizations avoid spending excessive resources on models that do not produce meaningful outcomes.

(McKinsey Global Institute, 2018)

精益人工智能开发

精益原则强调在开发过程中最大化价值，同时最小化浪费。

在人工智能项目中，精益开发侧重于：

- 快速原型化模型
- 使用小型数据集进行早期测试
- 消除不必要的复杂性
- 优先处理能带来业务价值的功能

这种方法帮助组织避免在无法产生有意义成果的模型上花费过多资源。

（麦肯锡全球研究院，2018年）

Minimum Viable AI Models

Similar to the concept of Minimum Viable Products (MVPs) in product development, AI projects often begin with Minimum Viable Models (MVMs).

Characteristics include:

- Basic predictive capability
- Limited dataset usage
- Fast development cycles
- Early stakeholder testing

The goal is not perfection, but learning whether the AI solution provides value before investing in full-scale deployment.

最小可行人工智能模型

类似于产品开发中的最小可行产品（MVP）概念，人工智能项目通常从最小可行模型（MVM）开始。

其特点包括：

- 基本的预测能力
- 有限的数据集使用
- 快速的开发周期
- 早期利益相关者测试

目标不是完美，而是在投入全面部署之前，了解人工智能解决方案是否能提供价值。

Hybrid Approaches: Agile + Traditional PM

Many organizations adopt hybrid project management approaches for AI initiatives.

Typical structure:

Strategic phases → Waterfall-like planning

Development phases → Agile experimentation

For example:

- Problem definition and governance → structured planning
- Model development → iterative Agile cycles
- Deployment and integration → structured rollout

This hybrid structure balances flexibility with organizational control.

混合方法：敏捷 + 传统项目管理

许多组织在人工智能项目中采用混合项目管理方法。

典型结构：

战略阶段 ⟹ 类似瀑布式的规划

开发阶段 ⟹ 敏捷实验

例如：

- 问题定义与治理 → 结构化规划
- 模型开发 → 迭代式敏捷周期
- 部署与集成 → 结构化上线

这种混合结构在灵活性与组织控制之间取得平衡。

Agile Challenges in AI Projects

Despite its advantages, Agile adoption in AI projects presents several challenges:

- Difficulty estimating timelines for experimentation
- Dependence on specialized technical talent
- Communication gaps between technical and business teams
- Managing expectations about model accuracy
- Integrating experimental workflows into enterprise governance

Project managers must therefore act as facilitators between technical teams and decision-makers.

人工智能项目中的敏捷挑战

尽管敏捷方法有其优势，但在人工智能项目中采用敏捷方法也面临一些挑战：

- 难以预估实验的时间线
- 对专业技术人才的依赖
- 技术团队与业务团队之间的沟通差距
- 管理关于模型准确性的期望
- 将实验性工作流程整合到企业治理中

因此，项目经理必须在技术团队和决策者之间扮演协调者的角色。

Key Takeaways

- AI development involves high uncertainty and experimentation
- Traditional Waterfall approaches struggle with AI workflows
- Agile methods enable iterative model development and rapid feedback
- Lean AI practices reduce wasted effort in model experimentation
- Many organizations implement hybrid PM structures for AI initiatives

Effective AI project management requires balancing agility with governance and strategic alignment.

关键要点

- 人工智能开发涉及高度的不确定性和实验性
- 传统的瀑布式方法难以适应人工智能工作流程
- 敏捷方法能够实现迭代式模型开发和快速反馈
- 精益人工智能实践减少了模型实验中的浪费
- 许多组织在人工智能项目中采用混合项目管理结构

有效的人工智能项目管理需要在敏捷性与治理和战略一致性之间取得平衡。

THANKS !!!



谢谢！



Welcome

COURSE:
ADVANCED PROJECT MANAGEMENT IN AI

Lesson 5 Session 2

Agile and Team Management in AI Projects:

Scrum in AI Projects

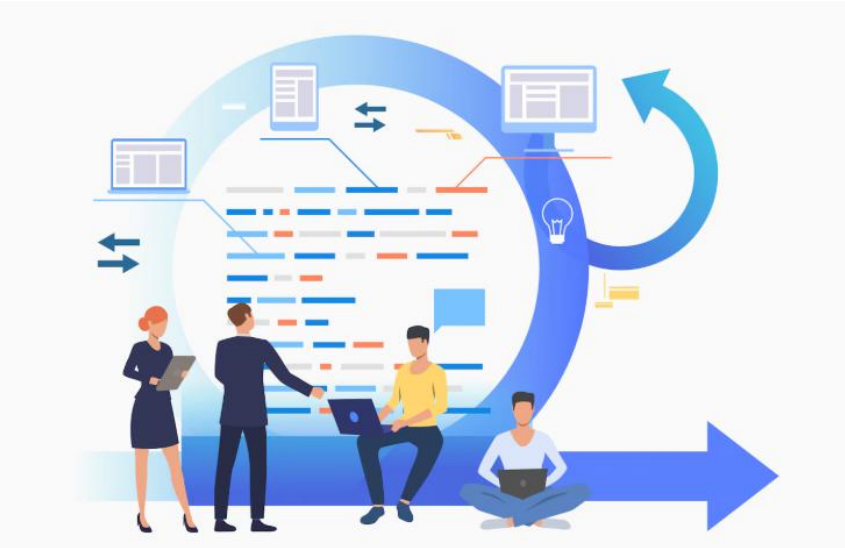
Welcome

课程：
人工智能高级项目管理

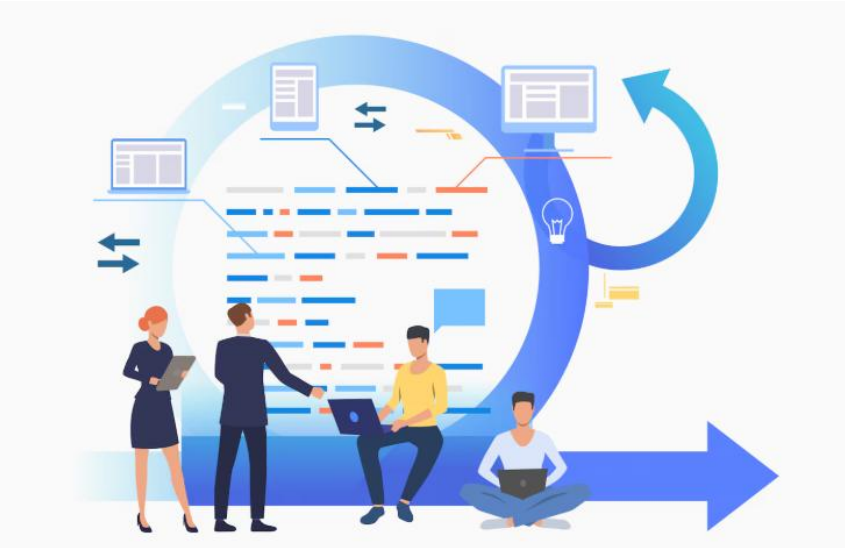
第5课 第2节

人工智能项目的敏捷与团队管理：
人工智能项目中的Scrum

LESSON 5.2 – Scrum in AI Projects



第5.2课 – 人工智能项目中的Scrum



Learning Objectives

By the end of this session, you should be able to:

- Explain how Scrum supports experimentation in AI projects
- Understand the roles of Product Owner, Scrum Master, and AI development teams
- Adapt sprint planning and backlog management for machine learning work
- Interpret sprint reviews using model performance metrics
- Apply Scrum practices while managing uncertainty in AI development

学习目标

在本部分课程结束时，您将能够：

- 解释Scrum如何支持人工智能项目中的实验
- 理解产品负责人、Scrum Master和人工智能开发团队的角色
- 为机器学习工作调整冲刺规划和待办事项列表管理
- 使用模型性能指标解读冲刺评审
- 在管理人工智能开发不确定性的同时应用Scrum实践

Why Scrum Matters for AI Project Managers

AI projects rarely progress in a straight line.

Unlike traditional software projects, AI development involves experimentation, data discovery, and iterative improvement.

For project managers, Scrum provides a structured framework to:

- Manage iterative experimentation cycles
- Deliver incremental model improvements
- Maintain visibility across technical and business stakeholders
- Reduce risk through frequent inspection and adaptation

Scrum allows teams to learn and adapt continuously while still maintaining delivery discipline.

为什么Scrum对人工智能项目经理很重要

人工智能项目很少直线推进。

与传统的软件项目不同，人工智能开发涉及实验、数据发现和迭代改进。

对于项目经理而言，Scrum提供了一个结构化的框架来：

- 管理迭代实验周期
- 交付增量式的模型改进
- 保持技术和业务利益相关者之间的可见性
- 通过频繁的检查和调整来降低风险

Scrum使团队能够在保持交付纪律的同时，持续学习和适应。

Overview of the Scrum Framework -1

Scrum is an Agile framework designed for complex and uncertain work environments.

Core characteristics include:

- Short, time-boxed iterations called sprints
- Continuous stakeholder feedback
- Self-organizing teams
- Incremental delivery of value

Scrum框架概述 - 1

Scrum是一种为复杂和不确定的工作环境设计的敏捷框架。

核心特征包括：

- 称为冲刺的短时、时间盒限定的迭代
- 持续的利益相关者反馈
- 自组织团队
- 价值的增量交付

Overview of the Scrum Framework -2

Typical Scrum elements include:

- Product Backlog
- Sprint Planning
- Sprint Execution
- Sprint Review
- Sprint Retrospective

These cycles allow teams to adjust direction as new insights emerge, which is essential in AI projects.

(Schwaber & Sutherland, 2020)

Scrum框架概述 - 2

典型的Scrum元素包括：

- 产品待办事项列表
- 冲刺规划
- 冲刺执行
- 冲刺评审
- 冲刺回顾

这些周期允许团队随着新见解的出现而调整方向，这在人工智能项目中至关重要。

（Schwaber & Sutherland，2020年）

Scrum Roles in AI Projects -1

Scrum defines three key roles:

Product Owner

Responsible for maximizing product value.

In AI projects this includes:

- Defining ML-driven business objectives
- Prioritizing model features
- Aligning technical work with organizational goals

Scrum Master

- Facilitates the Scrum process
- Removes obstacles affecting experimentation or infrastructure

人工智能项目中的Scrum角色 - 1

Scrum定义了三个关键角色：

产品负责人

负责最大化产品价值。

在人工智能项目中，这包括：

- 定义由机器学习驱动的业务目标
- 对模型功能进行优先级排序
- 使技术工作与组织目标保持一致

Scrum Master

- 促进Scrum流程
- 移除影响实验或基础设施的障碍

Scrum Roles in AI Projects -2

Scrum Master

- Facilitates the Scrum process
- Removes obstacles affecting experimentation or infrastructure

Development Team

In AI projects the team may include:

- Data scientists
- ML engineers
- Data engineers
- Domain experts

The team collaboratively delivers increments of model capability.

人工智能项目中的Scrum角色 - 2

Scrum Master

- 促进Scrum流程
- 移除影响实验或基础设施的障碍

开发团队

在人工智能项目中，团队可能包括：

- 数据科学家
- 机器学习工程师
- 数据工程师
- 领域专家

团队协作交付模型能力的增量。

AI Sprint Cycles

In AI projects, sprints typically focus on learning and model improvement rather than finished software features.

Typical sprint goals may include:

- Improving prediction accuracy
- Testing new algorithms
- Cleaning or expanding datasets
- Evaluating model robustness

Unlike traditional development, sprint outcomes may include:

- Experimental results
- Model prototypes
- Performance evaluations

The emphasis is on validated learning rather than code completion.

人工智能冲刺周期

在人工智能项目中，冲刺通常侧重于学习和模型改进，而非完成的软件功能。

典型的冲刺目标可能包括：

- 提高预测准确性
- 测试新算法
- 清理或扩展数据集
- 评估模型鲁棒性

与传统的开发不同，冲刺成果可能包括：

- 实验结果
- 模型原型
- 性能评估

重点在于验证性学习，而非代码完成。

Structuring Sprints for Machine Learning Work

AI teams often structure sprints around experimentation cycles.

A typical AI sprint may include:

1. Hypothesis definition
2. Data preparation
3. Model training
4. Evaluation and validation
5. Documentation of results

Key project management consideration:

Not every sprint will produce an improved model.

However, every sprint should produce insights that guide the next iteration.

为机器学习工作构建冲刺

人工智能团队通常围绕实验周期构建冲刺。

一个典型的人工智能冲刺可能包括：

1. 假设定义
2. 数据准备
3. 模型训练
4. 评估与验证
5. 结果记录

关键的项目管理考量：

并非每个冲刺都会产生改进的模型。

然而，每个冲刺都应产生指导下一轮迭代的见解。

Product Backlog for AI Projects

The product backlog represents all potential work items required to deliver an AI solution.

Examples of **backlog items** include:

- Data acquisition tasks
- Feature engineering improvements
- Model architecture experiments
- Integration with existing systems
- Evaluation against benchmark datasets

AI backlogs often include research tasks and exploratory experiments, which are less predictable than typical software features.

人工智能项目的产品待办事项列表

产品待办事项列表代表了交付人工智能解决方案所需的所有潜在工作项。

待办事项项示例包括：

- 数据获取任务
- 特征工程改进
- 模型架构实验
- 与现有系统的集成
- 针对基准数据集的评估

人工智能待办事项列表通常包含研究任务和探索性实验，其可预测性低于典型的软件功能。

Backlog Refinement for Machine Learning Features

Backlog refinement ensures upcoming tasks are clearly defined and feasible.

For AI work, refinement typically includes:

- Clarifying data requirements
- Defining evaluation metrics
- Identifying infrastructure needs
- Estimating experiment complexity

A well-refined backlog helps teams avoid ambiguous research goals that may delay progress.

Project managers should ensure each backlog item includes:

- A clear objective
- A measurable success criterion

针对机器学习功能的待办事项列表细化

待办事项列表细化的目的是确保即将开展的任务得到清晰定义且切实可行。

对于人工智能工作，细化通常包括：

- 明确数据需求
- 定义评估指标
- 确定基础设施需求
- 预估实验复杂性

一个经过良好细化的待办事项列表有助于团队避免可能延误进度的模糊研究目标。

项目经理应确保每个待办事项项都包含：

- 一个清晰的目标
- 一个可衡量的成功标准

Sprint Reviews with Model Metrics

Sprint reviews evaluate the outcomes of experimentation.

In AI projects, reviews often focus on model performance metrics, such as:

- Accuracy
- Precision and recall
- F1 score
- Latency or inference speed

Instead of asking “Was the feature completed?”, stakeholders evaluate:

“Did the experiment improve the model or generate useful insight?”

This ensures decisions remain evidence-based and data-driven.

(Goodfellow, Bengio, & Courville, 2016)

结合模型指标的冲刺评审

冲刺评审用于评估实验的结果。

在人工智能项目中，评审通常关注模型性能指标，例如：

- 准确率
- 精确率和召回率
- F1分数
- 延迟或推理速度

与其问"功能是否完成了？"，利益相关者评估的是：

"这个实验是改进了模型还是产生了有用的见解？"

这确保了决策始终基于证据和数据驱动。

（Goodfellow, Bengio, & Courville，2016年）

Managing Uncertainty in AI Sprints

AI development involves significant uncertainty due to:

- Data variability
- Model sensitivity to parameters
- Unexpected bias or overfitting

Scrum helps manage this uncertainty by:

- Limiting work to short sprint cycles
- Encouraging rapid experimentation
- Enabling quick course correction

Project managers must ensure stakeholders understand that AI progress may be incremental rather than linear.

管理人工智能项目冲刺中的不确定性

人工智能开发涉及显著的不确定性，其来源包括：

- 数据的可变性
- 模型对参数的敏感性
- 意外的偏差或过拟合问题

Scrum 框架通过以下方式帮助管理这种不确定性：

- 将工作限定在较短的冲刺周期内
- 鼓励快速实验
- 能够快速进行方向修正

项目管理者必须确保利益相关方理解，人工智能项目的进展可能是渐进式的，而非线性的。

Challenges of Applying Scrum to AI Projects

Despite its benefits, Scrum must be adapted carefully.

Common challenges include:

- Difficulty estimating ML experimentation tasks
- Dependency on large datasets or compute resources
- Interdisciplinary communication barriers
- Pressure to deliver results before models mature

Effective Scrum implementation requires flexible planning and strong communication across technical and business teams.

将 Scrum 应用于人工智能项目所面临的挑战

尽管 Scrum 具有诸多益处，但仍需谨慎调整以适应 AI 项目。

常见的挑战包括：

- 难以估算机器学习实验类任务
- 对大型数据集或计算资源的依赖
- 跨学科沟通障碍
- 在模型成熟前即面临交付成果的压力

有效的 Scrum 实施需要灵活的规划，以及技术团队与业务团队之间强有力的沟通。

Key Takeaways

Scrum provides an effective framework for managing AI projects because it:

- Supports iterative experimentation
- Enables continuous stakeholder feedback
- Encourages cross-functional collaboration
- Focuses on incremental learning and improvement

For project managers, the goal is not only to manage tasks but to create an environment where experimentation leads to measurable business value.

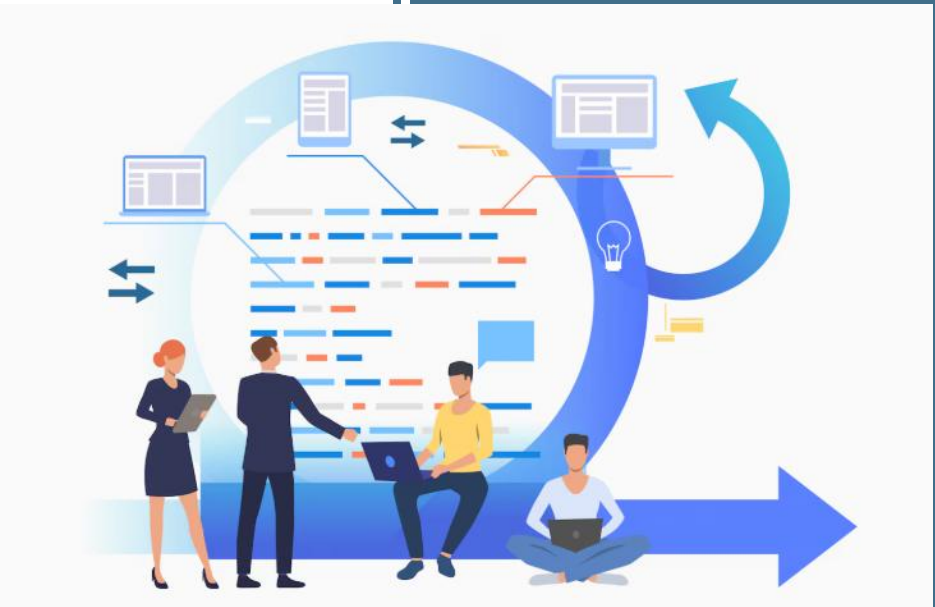
关键要点

Scrum 为管理人工智能项目提供了一个有效的框架，原因在于它：

- 支持迭代式实验
- 能够实现持续的利益相关方反馈
- 鼓励跨职能协作
- 聚焦于渐进式学习和持续改进

对于项目管理者而言，目标不仅在于管理任务，更在于营造一个能够让实验催生可衡量业务价值的环境。

THANKS !!!



谢谢！



Welcome

COURSE:
ADVANCED PROJECT MANAGEMENT IN AI

Lesson 5 Session 3

Agile and Team Management in AI Projects:
Communication and Stakeholder Management

Welcome

课程：
人工智能高级项目管理

第5课 第3节

人工智能项目的敏捷与团队管理：

沟通与利益相关方管理

LESSON 5.3 – Communication and Stakeholder Management in AI Projects



第5.3课 – 人工智能项目中的沟通与利益相关方管理



Learning Objectives - Lesson 5.3

- Explain why communication challenges are amplified in AI projects
- Identify how AI results can be misinterpreted by non-technical stakeholders
- Communicate model uncertainty and limitations effectively
- Translate AI outputs into business-relevant insights for executives
- Apply stakeholder management strategies in cross-disciplinary AI teams

学习目标 - 第5.3课

- 阐述为何人工智能项目中的沟通挑战会被放大
- 识别人工智能结果可能被非技术利益相关方误解的方式
- 有效沟通模型的不确定性与局限性
- 将人工智能输出转化为面向高管的业务洞察
- 在跨学科人工智能团队中应用利益相关方管理策略

Why This Topic Matters for Project Managers - 1

AI projects involve multiple professional domains working together:

- Business leaders defining strategic goals
- Data scientists developing models
- Engineers building infrastructure
- Legal and compliance teams addressing risk and ethics

为何此议题对项目管理者至关重要 - 1

人工智能项目涉及多个专业领域协同工作：

- 业务领导者负责界定战略目标
- 数据科学家负责开发模型
- 工程师负责构建基础设施
- 法务与合规团队负责应对风险与伦理问题

Why This Topic Matters for Project Managers - 2

Unlike traditional IT systems:

- AI outputs are probabilistic rather than deterministic
- Model behavior may not always be easily explainable
- Stakeholders often overestimate AI capabilities

For project managers, this creates a key responsibility: translating complex technical outcomes into clear, reliable business understanding.

Poor communication can lead to:

- unrealistic expectations
- mistrust in AI systems
- failed adoption despite technically successful models.

为何此议题对项目管理者至关重要 - 2

与传统IT系统不同：

- 人工智能输出具有概率性而非确定性
- 模型行为未必总是易于解释
- 利益相关方往往高估人工智能的能力

对项目管理者而言，这带来了一项关键职责：将复杂的技术成果转化为清晰、可靠的业务认知。

沟通不畅可能导致：

- 不切实际的预期
- 对人工智能系统的不信任
- 即便模型在技术上成功，也难以获得采纳

How AI Communication Gets Misinterpreted - 1

AI project communication often breaks down because technical language is interpreted differently by non-technical stakeholders.

Common misinterpretations include:

- Accuracy interpreted as certainty

Stakeholders may assume that a model with 90% accuracy will perform correctly in nearly every case.

- Predictions interpreted as guarantees

AI outputs may be perceived as definitive answers rather than probabilistic estimates.

人工智能沟通如何被误解 - 1

人工智能项目中的沟通常常受阻，原因在于非技术背景的利益相关方对技术语言存在不同解读。

常见的误解包括：

- 准确率被理解为确定性

利益相关方可能认为，一个准确率达90%的模型在几乎所有情况下都能正确运行。

- 预测被理解为保证

人工智能输出可能被视为确定的答案，而非概率性的估计。

How AI Communication Gets Misinterpreted - 2

- Technical confidence mistaken for reliability

Statements about model confidence may be interpreted as proof that the model is fully dependable.

- Iteration interpreted as project failure

Frequent experimentation cycles may appear to executives as lack of progress.

- Implication for project managers:

Communication must anticipate possible misunderstandings and clarify meaning proactively.

人工智能沟通如何被误解 - 2

- 技术置信度被误认为可靠性

关于模型置信度的表述，可能被解读为模型完全可靠的证明。

- 迭代被理解为项目失败

频繁的实验周期在管理者看来可能像是缺乏进展。

- 对项目管理者的启示：

沟通必须预判可能的误解，并主动加以阐明。

Communicating AI Uncertainty Effectively - 1

AI systems inherently involve uncertainty and probabilistic reasoning.

Project managers must ensure this uncertainty is communicated clearly and responsibly.

Effective communication practices include:

Careful use of probability language

Instead of: Model accuracy is 92%.

Explain:

In historical data, the model correctly predicted outcomes in 92% of cases, meaning errors remain possible

有效沟通人工智能的不确定性 - 1

人工智能系统本质上涉及不确定性和概率性推理。
项目管理者必须确保以清晰且负责任的方式传达这种不确定性。
有效的沟通实践包括：
审慎使用概率性语言
避免说：模型准确率为92%。
应解释为：
在历史数据中，该模型在92%的情况下正确预测了结果，这意味着错误仍然可能发生。

Communicating AI Uncertainty Effectively - 2

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Clarifying limitations

Stakeholders should understand situations where the model may perform poorly, such as:

- limited or biased data
- unusual cases outside the training dataset
- changing environments over time

Using scenario-based explanations

Example:

Out of 100 similar cases, the model may misclassify around eight cases. Uncertainty should be framed as a managed risk rather than a weakness.

有效沟通人工智能的不确定性 - 2

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阐明局限性

利益相关方应了解模型可能表现欠佳的情况，例如：

- 数据有限或存在偏差
- 训练数据集之外的异常情况
- 环境随时间变化

使用基于场景的解释

示例：

在100个相似案例中，模型可能错误分类约8个案例。不确定性应被界定为一种可管理的风险，而非模型的弱点。

Explaining Model Metrics to Non-Technical Stakeholders - 1

Machine learning teams rely on technical evaluation metrics that may not be meaningful to business audiences.

Common metrics include:

- Accuracy
- Precision
- Recall
- F1 score
- AUC-ROC

向非技术利益相关方解释模型指标 - 1

机器学习团队所依赖的技术评估指标，对业务方而言可能缺乏实际意义。

常见的指标包括：

- 准确率
- 精确率
- 召回率
- F1分数
- AUC-ROC值

Explaining Model Metrics to Non-Technical Stakeholders - 2

Project managers must translate metrics into operational meaning.

Examples:

Precision - proportion of flagged cases that are actually fraud

Recall - proportion of fraud cases the model successfully detects

False positives - additional investigations required by operations teams

The goal is to shift discussion from *technical statistics to business implications.*

(Sculley et al., 2015)

向非技术利益相关方解释模型指标 - 2

项目管理者必须将这些指标转化为具有业务含义的解释。

示例：

精确率 —— 在模型标记的案例中，实际为欺诈行为的比例

召回率 —— 模型成功检出的欺诈案例所占比例

假阳性 —— 需要运营团队进行额外核查的案例数量

目标在于将讨论焦点从*技术统计数据转向其对业务的实际影响。*

（Sculley 等人，2015）

Translating AI Outputs into Executive Insights - 1

Senior leaders are rarely interested in algorithmic details.

Executives typically want to understand:

- the business problem being solved
- the value created by the model
- the risks associated with relying on AI

将人工智能输出转化为高管洞察 - 1

高层领导通常对算法细节不感兴趣。

高管们通常希望了解：

- 所解决的业务问题是什么
- 模型创造了何种价值
- 依赖人工智能存在哪些风险

Translating AI Outputs into Executive Insights - 2

An effective communication structure includes:

1. Business objective
2. Key model insight
3. Operational impact
4. Risks and limitations

Example:

Customer churn prediction identifies 30% of potential churn cases two months earlier, enabling targeted retention campaigns. Communication should focus on decision support rather than algorithm design.

将人工智能输出转化为高管洞察 - 2

一种有效的沟通结构包括：

1. 业务目标
2. 关键模型洞察
3. 运营影响
4. 风险与局限性

示例：

客户流失预测模型可提前两个月识别出30%的潜在流失案例，从而能够开展有针对性的客户挽留活动。沟通应聚焦于决策支持，而非算法设计。

Managing Stakeholder Expectations in AI Projects

AI projects frequently experience challenges because stakeholders expect unrealistic performance.

Common misconceptions include:

- AI systems will achieve near-perfect accuracy
- AI will fully replace human decision-making
- once deployed, AI systems will remain stable indefinitely

Project managers must communicate early that:

- AI augments decision-making rather than replacing it
- performance may change due to data drift
- models require continuous monitoring and retraining

Expectation management should begin during project initiation and planning.

管理人工智能项目中的利益相关方期望

人工智能项目常遇挑战，原因在于利益相关方抱有不切实际的性能预期。

常见的误解包括：

- 人工智能系统将达到近乎完美的准确率
- 人工智能将完全取代人类决策
- 一旦部署，人工智能系统将永久保持稳定

项目管理者必须及早阐明：

- 人工智能是对决策的增强，而非替代
- 由于数据漂移，模型性能可能发生变化
- 模型需要持续的监控与再训练

期望管理应在项目启动与规划阶段即着手进行。

Stakeholder Mapping in AI Ecosystems - 1

AI projects often involve a broader set of stakeholders than traditional IT initiatives.

Typical stakeholder groups include:

- business leadership
- data science teams
- IT infrastructure teams
- legal and compliance departments
- customers and end users
- regulators and policymakers

人工智能生态系统中的利益相关方映射 - 1

相比传统IT项目，人工智能项目通常涉及更为广泛的利益相关方。

典型的利益相关方群体包括：

- 业务领导层
- 数据科学团队
- IT基础设施团队
- 法务与合规部门
- 客户与最终用户
- 监管机构与政策制定者

Stakeholder Mapping in AI Ecosystems - 2

Project managers can apply stakeholder analysis tools such as:

- power–interest matrices
- influence mapping
- communication planning frameworks

These tools help ensure the right information reaches the right stakeholders at the right time.

(Freeman, 1984)

人工智能生态系统中的利益相关方映射 - 2

项目管理者可运用利益相关方分析工具，例如：

- 权力-利益矩阵
- 影响力图谱
- 沟通规划框架

这些工具有助于确保恰当的信息在恰当的时机传达给恰当的利益相关方。

(Freeman, 1984)

Communication Tools for AI Projects - 1

Structured communication tools improve transparency and understanding in AI projects.

Examples include:

Model documentation

- Model cards describing model purpose and limitations
- Data sheets describing training datasets

人工智能项目的沟通工具 - 1

结构化的沟通工具能够提升人工智能项目的透明度和理解度。

示例包括：

模型文档

- 模型卡片：描述模型的目的与局限性
- 数据表：描述训练数据集

Communication Tools for AI Projects - 2

Visualization tools

- confusion matrices
- performance dashboards
- trend monitoring graphs

Project communication mechanisms

- sprint reviews focused on model outcomes
- documented model decisions and assumptions

These tools improve traceability, accountability, and shared understanding.

人工智能项目的沟通工具 - 2

可视化工具

- 混淆矩阵
- 性能仪表板
- 趋势监测图

项目沟通机制

- 聚焦于模型成果的冲刺评审
- 记录在案的模型决策与假设

这些工具有助于提升可追溯性、明确责任归属，并促进形成共同理解。

Conflict in Cross-Disciplinary AI Teams

AI teams combine experts with very different priorities and perspectives.

Typical tensions include:

Data scientists prioritizing model accuracy
Engineers prioritizing system scalability and reliability
Business leaders prioritizing speed and ROI
Compliance teams prioritizing risk mitigation and fairness

These priorities may conflict during project decision-making.

Project managers must balance:

- technical feasibility
- organizational objectives
- regulatory constraints

Effective communication becomes critical to align competing priorities.

跨学科人工智能团队中的冲突

人工智能团队汇聚了来自不同领域的专家，他们各自有着迥异的优先级和视角。

典型的紧张关系包括：

数据科学家优先考虑模型准确率
工程师优先考虑系统可扩展性与可靠性
业务领导者优先考虑速度与投资回报率
合规团队优先考虑风险缓解与公平性

在项目决策过程中，这些优先级可能相互冲突。

项目管理者必须平衡：

- 技术可行性
- 组织目标
- 监管约束

有效的沟通对于协调相互竞争的优先级至关重要。

Conflict Resolution Strategies for AI Teams

Project managers can apply structured approaches to resolve interdisciplinary conflicts.

Effective strategies include:

- Clarifying shared project goals: Ensure all stakeholders understand the primary business objective.
- Evidence-based decision making: Use experiments, metrics, and pilot results to guide discussions.
- Facilitated dialogue: Encourage communication between technical experts and business leaders.
- Managing trade-offs: Accept that improvements in accuracy may increase cost, complexity, or deployment time.

Successful AI projects often require balanced compromises rather than perfect solutions.

人工智能团队的冲突解决策略

项目管理者可运用结构化方法来化解跨学科冲突。

有效的策略包括：

- 明确共同的项目目标：确保所有利益相关方理解核心业务目标。
- 基于证据的决策：运用实验、指标和试点结果来引导讨论。
- 引导式对话：鼓励技术专家与业务领导者之间的沟通。
- 管理权衡取舍：认识到准确率的提升可能带来成本、复杂性或部署时间的增加。

成功的人工智能项目往往需要达成均衡的折衷方案，而非追求完美的解决方案。

Key Takeaways

Communication plays a strategic role in AI project success.

Key insights from this session:

- AI outputs are probabilistic and require careful explanation
- technical metrics must be translated into business meaning
- stakeholder expectations must be actively managed
- AI projects involve complex multi-stakeholder ecosystems
- project managers must facilitate communication across disciplines

Effective communication significantly increases AI adoption, trust, and project success.

关键要点

沟通在人工智能项目成功中发挥着战略性作用。

本节的核心见解包括：

- 人工智能输出具有概率性，需加以审慎解释
- 技术指标必须转化为业务层面的含义
- 利益相关方的期望必须得到主动管理
- 人工智能项目涉及复杂的多利益相关方生态系统
- 项目管理者必须促进跨学科的沟通

有效的沟通能够显著提升人工智能的采纳度、信任度以及项目成功率。

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THANKS !!!



谢谢！

